

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

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|---|------------------------------|
| Perfomax
<i>AI Advisor</i> | May 2026–present |
| National University of Singapore
<i>Postdoc under Kostya Novoselov</i> | 2022–present |
| <ul style="list-style-type: none">▪ Leadership :<ul style="list-style-type: none">· Program Chair for the AI4X 2025/6 conferences.· Main organizer of the ICLR 2025 workshop on multiscale machine learning.▪ Transformer for generating symmetric crystals :<ul style="list-style-type: none">· Created a generative model for material design based on symmetry inductive bias.· Implemented the model in PyTorch.· Achieved the best generation quality and diversity among space-group-conditioned models.· Led a team of 6.▪ ML for predicting properties of defects in 2D crystals :<ul style="list-style-type: none">· Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.· Developed code for reproducible parallel running of machine learning experiments.· Led a team of 3. | |
| Constructor Group
<i>Machine Learning Consultant</i> | February 2021–present |
| <ul style="list-style-type: none">▪ AI/ML Consulting | |
| CERN [Yandex → HSE University]
<i>Researcher</i> | 2014–2022 |
| <p>2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.</p> <ul style="list-style-type: none">▪ Muon identification at the LHCb experiment at CERN :<ul style="list-style-type: none">· Solved a classification problem over tabular data with noisy labels under timing constraints.· Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.· Integrated the model into LHCb production, mostly in C++.· Packaged the work as a data science competition problem for IDAO-2019.▪ CatBoost aka fighting biases with dynamic boosting :<ul style="list-style-type: none">· Set up a distributed system for running experiments for the team with Bash and Python.· Studied gradient boosting improvements with experiments on toy data. | |

Education

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| Higher School of Economics (HSE University)
PhD in Computer Science, supervisor Andrey Ustyuzhanin ; Machine Learning for particle identification in the LHCb detector . | 2016–2020 |
| Sapienza — Universita di Roma
PhD in Physics (double degree with HSE), supervisor Barbara Sciascia . | 2016–2020 |
| Product Management course at Yandex
Focused coursework in product strategy and execution. | Spring 2018 |
| Yandex School of Data Analysis
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems. | 2013–2015 |

Moscow Institute of Physics and Technology

MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.

BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 20).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a [science slam](#), interviews (1, 2), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in 2015, 2016, 2017, 2018, 2019, 2020, 2021, and 2023.
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Taught and developed an introductory data analysis [course](#) at HSE.
- Delivered more than 20 [conference talks](#).

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

ML & AI (Structured, non-text, domain-specific data) ·
Python (Research coding) ·
PyTorch (Deep learning) ·
Linux (Administration) ·
Research Writing & Speaking ·
Leadership & Mentorship

Selected Publications

1. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. [Position: Align AI to Our Aspirations, Not Our Flaws](#). Pluralistic Alignment Workshop at ICML 2026, June 2026.
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budenny, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). ICML 2026 AI for Science Workshop, May 2026.
3. Jose Recatala-Gomez, **Nikita Kazeev**, et al. [Generative design of inorganic materials](#). arXiv preprint arXiv:2604.14082, April 2026.
4. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
5. Stephen G. Dale, **Nikita Kazeev**, et al. [AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions](#). arXiv preprint arXiv:2511.20976, November 2025.
6. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
7. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).